

# Predation rates based on dummy caterpillars

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December 14, 2023

## 3 Week 05 ( November 30 - December 14 )

Find 3 datasets from a domain specific data portal/repository.

<https://www.bexis.uni-jena.de/ddm/data/Showdata/31113>

<https://www.bexis.uni-jena.de/ddm/data/Showdata/24246>

<https://www.bexis.uni-jena.de/ddm/data/Showdata/25807>

### What are the different types of errors in collected data?

- **Missing Data:**

There are entries with "NM" (Not Measured) in various columns. Total 125 in the first dataset, 280 in the second dataset, and 15 in the third dataset.

- **Incorrect Data Types:**

All the data types match the expected types for each column in each dataset. Deploy different types of filters on columns based on value types.

- **Outliers:**

There are no outliers in all of our three datasets, Deploy different types of filters on columns based on value types.

- **Inconsistencies:**

Check for inconsistencies in data entries, such as conflicting information between columns.

- In the first step, we install Pandas
- After that we load our data to Pandas data frame.(we have CSV file format )
- We also perform data analysis and check columns. Such as checking for negative values in columns, etc.

### How to Quantify or detect errors in collected data?

- **Descriptive Statistics:**

- Descriptive statistics is a branch of statistics that deals with the collection, analysis, interpretation, presentation, and organization of data. In Python, the pandas library is commonly used for data analysis, and it provides a describe() function that can be used to generate descriptive statistics of a data frame or Series.

#### • Data Profiling:

- We used Python libraries to make data profiling ( pandas-profiling )  
pandas-profiling is a Python library that generates profile reports from pandas DataFrame. This report includes information about the distribution of the variables, missing values, correlation, and more.

- At first, we installed pandas-profiling
- Load the dataset(In CSV format)
- After that, we Generate a profile report
- Save the report to an HTML file and Display the report in a Jupyter Notebook.
- We also attached an extra file for the part with the name **PandasProfilingReport.pdf**

#### • Visualization:

- We use Matplotlib to show our analysis plots.

#### • Functional Dependency:

First dataset:

- Start\_date → End\_date
- PlotID → Bexis\_PlotID

Second dataset:

- PlotID → PlotID2
- PlotID2 → Position

Third dataset:

- Date\_start → Date\_end
- Time\_start → Time\_end
- Nr\_dummies → Arthropod\_prop
- Nr\_dummies → Bird\_prop
- Nr\_dummies → Mammal\_prop
- Nr\_dummies → Gastropod\_prop
- Nr\_dummies → Predation\_prop

**How do we correct these errors?**

- **Imputation:**

In the first dataset, the only rows with NA values have the Dummy\_missing value equals to 1 ( it means that the dummy couldn't be retrieved from the field ) so we replaced all the NA values in that row with 0 because it shows that we were not able to observe any marks from any predators (birds, mammals and ...)

In the second dataset, we replaced NA value with 0 because it only can be 0 or 1 since it's binomial, and the default value for this mark is zero ( when there's no marks yet ) so the best replacement would be zero.

In the third dataset, we replaced the NA value with 0 because there are zero dummies in these rows as can be seen in the Nr\_dummies column, therefore there will be no fraction (since we have zero dummies and any fraction of zero equals zero)

- **Data Type Conversion:**

There is no need to convert the format of data because the value and data type are completely matched.

- **Outlier Handling:**

We didn't have any outliers in our three datasets.

- **Data Cleaning Algorithms:**

Since our datasets are clean there is no need to implement a cleaning algorithm.

## **Contributions**

### **Shirin Shoghli**

Searching for different types of errors in datasets (including Missing Data, Incorrect Data Types, Outliers, and Inconsistencies), detecting and correcting them applying filtering on Excel columns as well as performing Python libraries, visualizing data analysis plots

### **Shahrzad Torabi**

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### **Sanaz Bayat**

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### **Maryam Gheibi**

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### **Mozhdeh Ramezani Dastjerdi**

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